



QUIMICA

THROUGH CHEMICAL
FIND YOUR WAY



2024

NEWS LETTER

CHEMICAL ENGINEERING DEPARTMENT

TABLE OF CONTENTS



1 Director's Message

2 HOD's Message

3 Prof. In-Charge's Message

4 About Quimica

5 Our Department

6 Program Educational Objectives (PEOs)

7 Program Specific Objectives (PSOs)

8 Features

9 Lab Facilities

10 Faculty's Profile

11 Faculty's Message

12 Faculty's Article

13 Student's Research Articles

14 Student's Blogs

15 Our Achievers

16 Placement Report

17 Webinars, Workshops and Seminars

18 Recent Events

19 Alumni's Article

20 Recent Advancements in Chemical Industry

21 Noble Prize in Chemistry

22 Gallery

DIRECTOR'S MESSAGE



The Department of Chemical Engineering at BIT Sindri, established in 1956, is one of the institute's oldest and most prestigious academic departments. Renowned as a leading hub for Chemical Engineering in India, it has consistently garnered respect from both industries and academia. Since its inception, the department has worked tirelessly to nurture students and elevate their academic and professional standards.

With state-of-the-art facilities and dedicated faculty, the department empowers students to broaden their scientific horizons and develop a strong foundation in chemical engineering principles. Over the years, our students have excelled in various domains, securing positions in reputed organizations, pursuing internships at premier institutions like IITs, and exploring global opportunities. Our proud alumni continue to make significant contributions worldwide.

The motto, "Hold the Vision, Trust the Process," perfectly encapsulates the journey of the Chemical Engineering Department. Inspired by this philosophy, the department initiated Quimica, a dynamic platform to educate and inspire students. Quimica aims to provide insights into the diverse facets of chemical engineering through engaging blogs and articles, keeping students informed about the latest global advancements.

We extend our heartfelt congratulations to Team Quimica and the Department of Chemical Engineering for their visionary initiative, fostering a culture of knowledge-sharing and innovation. May Quimica continue to enlighten and inspire young minds for years to come.

HOD'S MESSAGE



"Life is about discovering your potential and utilizing it to the fullest," a wise man once said, and this philosophy resonates deeply with the Chemical Engineering family at BIT Sindri.

Established in 1956, the Department of Chemical Engineering is a hub of academic excellence, offering specialized B. Tech and M. Tech programs. With state-of-the-art laboratories and a focus on holistic development, the department prepares students to compete globally.

Our students excel in diverse sectors like R&D, consultancy, and PSUs, with alumni thriving in organizations such as IOCL, GAIL, ONGC, and Vedanta. Internships at CSIR-CIMFR, IIT-ISM Dhanbad, and IOCL, along with training at Tata Steel and Reliance, provide invaluable experience.

The department celebrates stellar academic achievements, with one student securing an impressive AIR 50 in GATE 2024. We are also proud of our NBA accreditation, which affirms the quality of our programs.

Quimica, our flagship initiative, fosters creativity, knowledge sharing, and innovation. Through a vibrant student community and a strong alumni network, Quimica keeps students informed about the latest advancements in chemical engineering.

On behalf of the Chemical Engineering Department, I extend my best wishes to Team Quimica for their continued success and impactful contributions!



DR. ASHOK KUMAR BARANWAL

"QUIMICA - Bridging the Gap Between Faculties and Students"

It gives me immense pride and joy to announce the release of Quimica's annual magazine. Publications like these play a vital role in nurturing and enhancing the writing and editorial skills of students, providing a platform for creative and intellectual expression.

The Association of Chemical Engineers (Quimica) has always served as an active bridge between the faculties and students of the department. Through its efforts, Quimica has fostered harmony and collaboration, driving the department toward greater heights. The dedication and enthusiasm of its members deserve recognition, as they continue to make Quimica one of the most vibrant and active societies at BIT Sindri.

I extend my heartfelt congratulations to the Quimica team for their initiative and dedication to this edition of the magazine. In a constantly evolving field like chemical engineering, it is crucial to reflect on our work and explore its relevance. This magazine serves as a medium to articulate these thoughts and rediscover the essence of our discipline.

I encourage students to contribute innovative articles with fresh perspectives for future editions. At Quimica, we provide an invaluable opportunity for self-realization and growth—a privilege we are proud to offer. Together, let's continue to break new ground and propel the Department of Chemical Engineering at BIT Sindri toward new heights.



DR. NIRUPAMA

"QUIMICA – A Platform to Unleash Creativity"

It is with great pride that I write this message for Quimica's annual magazine at BIT Sindri, a platform that brilliantly showcases students' creativity, originality, and imagination. The magazine reflects their innovative spirit and BIT Sindri's balanced approach to education, blending academic excellence with co-curricular and extracurricular pursuits. My heartfelt congratulations to the Director, faculty, and students for their dedication, which has elevated this institution to a renowned center of learning. I am especially grateful to Dr. Amit K. Gupta, our esteemed Head, for entrusting me with the privilege of leading Quimica.

In today's fast-evolving world, traditional teaching methods alone are insufficient. Quimica provides students with a platform to explore innovative ideas, develop critical skills like management and communication, and present their thoughts effectively. It is particularly beneficial for chemical engineering students, equipping them with essential tools for future success. To further enhance its impact, I suggest that future editions focus more on emerging industrial technologies and multidisciplinary topics, addressing real-world challenges to better prepare students for dynamic careers.

Quimica's success owes much to the enthusiastic participation of students and the unwavering support of the management, faculty, and staff. Their combined efforts have made BIT Sindri a hub of academic and research excellence. I extend my best wishes to the college and all its stakeholders for continued success and growth in the years to come.



ABOUT QUIMICA

Quimica is a dynamic society of the Chemical Engineering Department, dedicated to fostering innovation, skill development, and industry engagement. It organizes the annual technical colloquium in association with IICHE and AICHE, uniting students, alumni, and professors. By hosting blogs, webinars, and interactive events, Quimica provides a bridge between academia and industry, offering valuable learning opportunities.

The society aims to create a platform for aspiring engineers to acquire essential skills, explore advancements in chemical engineering, and contribute to impactful research. It supports initiatives like consultancy services for industries and entrepreneurs, promoting sustainable growth and innovation. Guided by dedicated faculty, Quimica has achieved remarkable milestones, becoming a center of academic and industrial excellence.

Through insightful webinars featuring renowned experts and flagship events like ElementaX and Jeopardy, Quimica delivers a holistic learning experience. It continues to inspire future engineers, preparing them to tackle real-world challenges in chemical engineering.

Quimica also prioritizes building a strong sense of community among its members by encouraging collaboration and teamwork. With its vibrant network of alumni, the society fosters mentorship opportunities, enabling students to gain valuable guidance and insights into their careers.



OUR DEPARTMENT



The Department of Chemical Engineering, founded in 1956, stands as one of the pioneering disciplines at BIT Sindri. It is widely recognized as a leading center for Chemical Engineering education in India by academia and industry alike.

Offering a four-year B.Tech. degree and an M.Tech. in Chemical Plant Design, the department boasts a team of experienced faculty actively engaged in industrial collaborations to drive innovation and research.

Equipped with advanced laboratories like the Unit Operations Lab, Process Control Lab, Petroleum Refinery Engineering Lab, Plastics Technology Lab, Process Engineering Lab, Environmental Engineering Lab and Chemical Engineering Thermodynamics Lab, the department ensures practical learning. Its global alumni network and skilled students further enhance its reputation.

MISSION STATEMENT



To impart quality technical education to UG and PG courses in Chemical Engineering.

To activate and pursue research in thrust/emerging areas of technology in Chemical Engineering.

To provide consultancy services to Industries and Entrepreneurs.

To create qualified human resources to cater the needs of a sound national economy through developmental activities.

To make the department a center of excellence for research and development in Chemical Engineering and related fields.

Program-Educational Outcomes



PEO1: Knowledge:

To generate high-quality humans with fundamental knowledge resources in our core areas of Chemical Engineering competence and the emerging fields of research, as well.

PEO2: Design Abilities:

To make a valuable contribution to technology with problem-solving skills, laboratory skills, and design skills for technical careers in solving critical problems for the social and economic development of the nation.

PEO3: Professional Concern:

To exert focused efforts to generate graduate students, competent for finding their place in leading companies and work as an effective team members with communication and teamwork skills as well as an appreciation for ethical behavior necessary to thrive in their careers.

PEO4: Specialization aspects:

Graduates of the program will be prepared to pursue career choices in chemical engineering, allied and interdisciplinary fields of industries and renowned educational institutes for pursuing higher education that benefit from a strong background in applied sciences or engineering.

PEO5: Self-Learning:

Graduates will be practiced to persist their professional development through education and personal development established on their awareness of library resources and professional societies, workshops, industrial visits, expert talks, industry interactions, etc.

PROGRAM SPECIFIC OBJECTIVES

PSO1: Apply the principles of mathematics, science, and foundational chemical engineering knowledge to address complex challenges, solve critical issues, and design equipment for core chemical and allied industries.

PSO2: Develop new process plants by analyzing basic requirements and conducting feasibility studies. Utilize expertise in instrumentation, process dynamics and control, and process design. Leverage modern chemical engineering tools such as CHEMCAD, MATLAB, and others for modeling and simulation.

PSO3: Demonstrate readiness to work in chemical industries by identifying and addressing health, safety, legal, environmental, and cultural aspects. Excel in professional careers across chemical, petroleum, petrochemical, pharmaceutical, food, biotechnology, energy, materials processing, and other related industries and organizations.



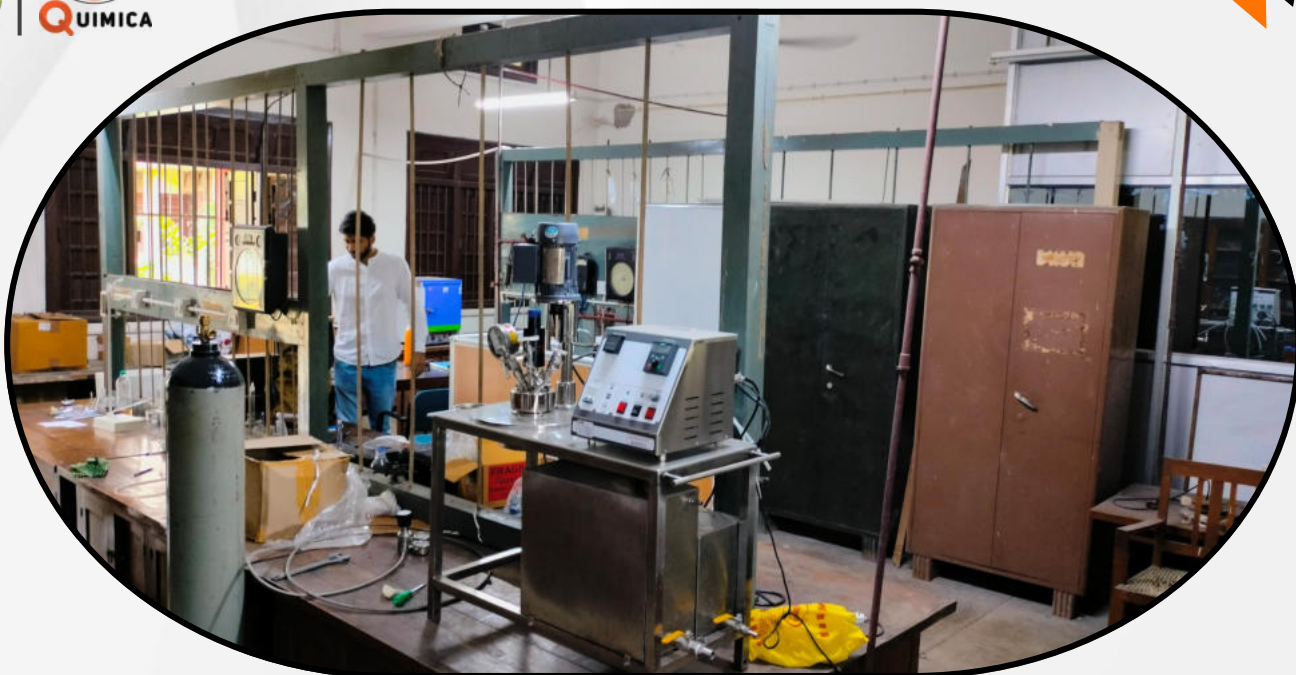
Features

The department admits 120 students annually for the B.Tech. program and 25 students for the M.Tech. program.

Students are trained in Computer-Aided Design during the 7th semester to enhance computational skills for modern industrial needs.

Highly qualified faculty members mentor students in B.Tech. and M.Tech. projects, fostering their research aptitude and analytical skills.

The department organizes industrial tours to organisation such as SAIL, IOCL, HINDALCO, ACC, CFRI, and BCCL, providing students with valuable industrial exposure and insights.



LAB FACILITIES

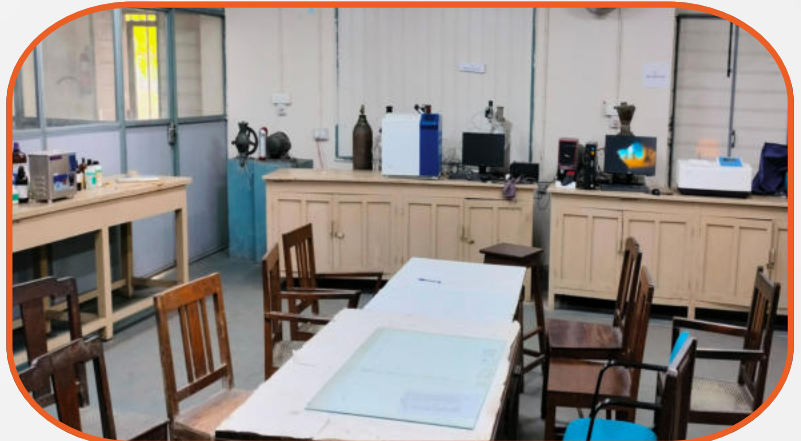
- Fluid Mechanics Lab
- Fluidization Engineering Lab
- Thermodynamics Lab
- Process Control Lab
- Petroleum Refinery Lab
- Process Engineering Lab
- Unit Operations Lab
- Characterization Lab





PCE Lab

Characterisation Lab



Mass Transfer
Operations Lab



Fluidization Lab



Thermodynamics Lab



Environmental Lab





FACULTY'S PROFILE



Dr. Amit Kumar Gupta

H.O.D & Associate Professor

Academic Background :-

PhD in Chemical Engineering, IIT Delhi

M-Tech in Chemical Engineering, IIT Bombay

B-Tech in Chemical Engineering, HBTI Kanpur

Research Area :-

Heat Transfer, Computational Fluid Mechanics, Interfacial Science

Dr. Amar Kumar

Assistant Professor

Academic Background :-

PhD in Chemical Engineering, VBU Hazaribagh

M-Tech in Chemical Engineering, BIT SINDRI

B-Tech in Chemical Engineering, BIT SINDRI

Research Area :-

Computational Fluid Mechanics, Fluidization Engineering



Dr. Ajay Oraon

Assistant Professor

Academic Background :-

PhD in Chemical Engineering, IIT-ISM Dhanbad

M-Tech in Chemical Engineering, IISc Bangalore

B-Tech in Chemical Engineering, BIT Sindri

Research Area :-

Fluid Mechanics, Separation Process



FACULTY'S PROFILE



Dr. Usha Kumari

Assistant Professor

Academic Background :-

PhD in Chemical Engineering, IIT Kharagpur
M-Tech in Chemical Engineering, IIT Roorkee
B-Tech in Chemical Engineering, BIT Sindri

Research Area :-

Pollution Control, Material Surface Engineering, Biomass, Pyrolysis

Dr. Sunil Kumar Singh

Assistant Professor

Academic Background :-

PhD in Chemical Engineering, IIT Guwahati
M-Tech in Chemical Engineering, IIT Kanpur
B-Tech in Chemical Engineering, BIT Sindri

Research Area :-

Micro and Nano Fluidics, Biosensors, Colloid and interface science



Prof. Manish Kumar

Assistant Professor

Academic Background :-

M-Tech in Chemical Engineering, IIT Roorkee
B-Tech in Chemical Engineering, BIT Sindri

Research Area :-

Computational Fluid Dynamics, Heat Transfer, Fluid Mechanics



FACULTY'S PROFILE



Dr. Diwakar Pandey

Assistant Professor

Academic Background :-

PhD in Chemical Engineering, IIT-BHU Varanasi
M-Tech in Chemical Engineering, IIT-BHU Varanasi
B-Tech in Chemical Engineering, B.I.E.T. Jhansi

Research Area :-

CO₂ capture, Bio-gas purification, Industrial Pollution Control

Dr. Abhishek Anand Hembrom

Assistant Professor

Academic Background :-

PhD in Mineral Engineering, IIT-ISM Dhanbad
M-Tech in Mineral Engineering, IIT-ISM Dhanbad
B-Tech in Chemical Engineering, BIT Sindri

Research Area :-

Coal & Mineral processing & its Beneficiation, Coal Preparation, etc.



Dr. Ashok Kumar Baranwal

Assistant Professor

Academic Background :-

PhD in Chemical Engineering, IIT Kanpur
M-Tech in Petroleum Engineering, IIT-ISM Dhanbad
B.Sc. in Chemical Engineering, BIT Sindri

Research Area :-

Rheology of Complex Fluids, Modeling of Non-Newtonian Fluids, Computational Fluid Dynamics



FACULTY'S PROFILE



Dr. Nirupama

Assistant Professor

Academic Background :-

PhD in Chemical Engineering, IIT Roorkee

M-Tech in Chemical Engineering, IIT Roorkee

B-Tech in Chemical Engineering, Punjab Technical University

Research Area :-

Polymer composites, Risk assessment

Prof. Pitho Hansda

Assistant Professor

Academic Background :-

M-Tech in Mineral Engineering, IIT-ISM Dhanbad

B-Tech in Chemical Engineering, BIT, Sindri

Research Area :-

Mineral Beneficiation, Chemical Technology



Dr. Ch. V. Raghunath

Assistant Professor

Academic Background :-

PhD in Chemical Engineering, IIT-BHU Varanasi

M-Tech in Chemical Engineering, SRM university, Chennai

B-Tech in Chemical Engineering, A.N University

Research Area :-

Air pollution control, Multi component absorption, Process Development



FACULTY'S PROFILE



Prof. (Ms.) Devina Ratnam

Assistant Professor

Academic Background :-

PhD (Pursuing) in Chemical Engineering, IIT-ISM Dhanbad

M-Tech in Chemical Engineering, IIT Gandhinagar

B-Tech in Chemical Engineering, BIT Sindri

Research Area :-

Nanotechnology synthesis of boron based nanosheets

Prof. (Ms.) Poornima Pandey

Assistant Professor

Academic Background :-

PhD (Pursuing) in Chemical Engineering, IIT (BHU) Varanasi

M-Tech in Chemical Engineering, IIT (BHU) Varanasi

B-Tech in Chemical Engineering, UP Technical university

Research Area :-

Air pollution control, Catalytic Separation Processes





FACULTY'S MESSAGE



Dr. Amar Kumar
Assistant Professor,
Chemical Engineering Department,
B.I.T. Sindri

"Quimica: Shaping Chemical Engineering's Future with Knowledge and Passion."

The fact that Quimica is renewing its electronic publication for the fifth edition on behalf of the Department of Chemical Engineering at BIT Sindri, Dhanbad, is a joyful occasion. For years, Quimica has allowed our students to learn and exhibit various technical and soft skills. The Department and Quimica have made every effort to provide budding engineers with useful information and insightful direction, not just in the area of chemical engineering but also in a wider sense. Our students have made a significant influence on the national as well as international levels. In numerous national-level competitive exams like the GATE and CAT, many of our students have once again performed exceptionally well. This year, one of our chemical engineering students were awarded the esteemed MITACS scholarship and completed their three-month research internships at various renowned Canadian institutions.



FACULTY'S MESSAGE



Prof. Manish Kumar
Assistant Professor,
Chemical Engineering Department,
B.I.T. Sindri

"Empowering Chemical Engineering's Future Through Quimica's Commitment."

To foster a culture of all-round student development at the Department of Chemical Engineering so that graduates may compete in the global marketplace.

The valuable resource that all teachers have is each other. Without cooperation, development is limited by my perspective. The student is energetic while the teacher is experienced. As a teacher, it is our duty to channelize the energy of the students in the right direction by sharing our experience and knowledge. So that the students keep establishing new dimensions in life by making efficient use of the experience and knowledge with their hard work and energy.

All of the students' enthusiastic engagement and positive feedback are much appreciated. Special thanks to the Quimica team, without whom this wouldn't be possible.

FACULTY'S ARTICLE

Sulphur Removal Plant from Naphtha



Dr. Amit Kumar Gupta
Associate Professor, B.I.T. Sindri

Primary Sulphur Removal Plant:-

Sweet naphtha is processed naphtha with a Sulphur content of less than 50 ppm and sour naphtha is processed naphtha with a Sulphur content of more than 90 ppm.

- Sour naphtha is handled in the principal Sulphur Removal Plant.
- Sweet naphtha is simply stored in overhead tanks.
This naphtha is now delivered to the vaporizer.

Vaporizer and hydrodesulfurizer (HDS) :-

- A mild steel vessel with refractory connecting serves as the vaporizer. Syngas is combined with the processed naphtha from the PSR plant before heating it from outside. Cr-Mo spiral tubes operate below 625°C, with effluent temperature crucial to the unit's performance.
 - This is managed by maintaining fuel flow at a constant pressure.
- A basic calculation uses a temperature of 50°C, a pressure of 35 kg/cm² and a pressure drop of 2 kg/cm².
- The HDS unit is where the vaporized process naphtha goes to get its Sulphur removed. The catalyst must be manually inserted into the holding grids. It consists of a purple sandwich with two ZnO beds surrounding a CO-MO bed.

The rapid ZnO bed, which is created by the reaction of hydrogen and reactive Sulphur in the processed naphtha, removes the H₂S produced by the reaction between hydrogen and non-reactive Sulphur in the presence of CO-MO catalysis. When the initial ZnO bed reaches saturation, the ability to change the H₂S inlet and outflow is offered and the bed is turned around.

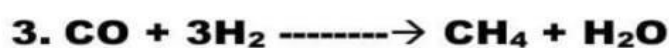
The HDS should not be heated above 390°C to prevent the catalyst from sintering and the generation of the unfavourable sooty carbon. On the other hand, the temperature at the inlet shouldn't be too low to prevent the need for additional fuel to heat the stream going to the main reformer.

The process naphtha from the HDS is combined with steam before going into the primary reformer. For the steam-carbon ratio, the steam flow rate is roughly five times that of naphtha. The steam-carbon ratio is designed to be around 3.5.

The process naphtha from the HDS is combined with steam before going into the primary reformer. For the steam-carbon ratio, the steam flow rate is roughly five times that of naphtha. The steam-carbon ratio is designed to be around 3.5.

Primary Reformer:

In the primary reformer, the reforming of the process of naphtha occurs. Major reactions involved here are:



All of the above reactions are endothermic.

In the primary reformer, there are four rows of 33 tubes each, i.e., 132 tubes in all. The tubes are made of HKO, an alloy of Cr, Ni, and Ti additives. Pigtails are provided at the inlet and outlet of the primary reformer to release thermal stresses. The catalyst filled inside the tubes is a Ni catalyst with a ceramic carrier (alumina). No promoter is used. Fuel naphtha is burnt to generate heat in the primary reformer.

Shift Converter and Carbon Dioxide Removal:

The gas stream coming from the MGG contains CO_2 , CO, H_2 , N_2 , and other inserts. CO and CO_2 should be removed before entering into the synthesis loop due to their poisonous nature. The removal of CO is inefficient compared to the removal of CO_2 . Hence CO is first converted to CO_2 and then removed. Any traces of CO left in the process stream is consumed by hydrogen in the Methanator step.



FACULTY'S ARTICLE

COAL TAR DISTILLATION



Dr. Amar Kumar

Assistant Professor, B.I.T. Sindri

The coal tar produced by coking coal is a dark, viscous liquid with distinctively foul odours. It was once seen as a nuisance in the production of coal gas in addition to being a waste product. However, it was discovered to be a secret treasure of numerous aromatic compounds that are currently used as the raw material for lovely dyes, medications, explosives, photographic developers, goods and many other items that are used in our everyday lives.

COAL TAR DISTILLATION:

At least 200 aromatic compounds can be found in coal tar, but the following are the most significant ones:

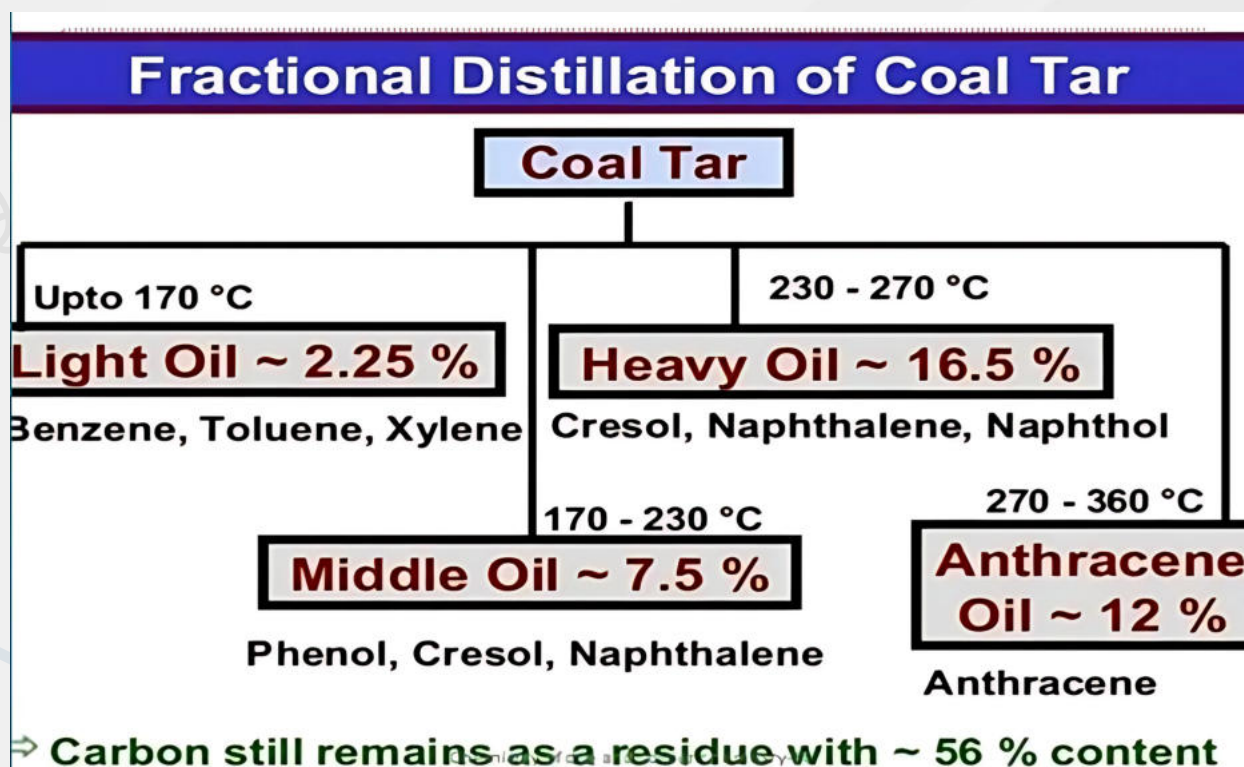
- Benzene, toluene, xylene, naphthalene, anthracene and other hydrocarbons.
- Xylenols, cresols and other phenols.
- Quinoline and Pyridine are basic nitrogen compounds.

The fractional distillation process separates the components from coal tar. Following the removal, the real distillation is done in a sizable wrought-iron still with a steam pipe installed within that is heated below by fire. Connected to the still head is a condensing worm that is submerged in water.

Process steam is introduced into the still towards the end of the steam to prevent the residue from caking. Condensed goods are transported into various receivers from the heater. The table below lists the cuts made to the typical technical distillation of tar:

Fraction	Distillation Temp (°C)	SP.	Chief Constituents
Light Oil	Up to 170 °C	0.9	Benzene, Toluene, Xylene, Naphthalene, Anthracene
Middle Oil	170-230	1.0 to 1.02	Phenols, Naphthalene
Heavy Oil	230-270	1.02 to 1.1	Cresols, Phenols
Anthracene Oil	270-360	1.1	Anthracene

Each fraction is worked and separated into its constituents:



1. First Fraction (Light Oil, up to 1700C):

It has benzene, toluene, and xylenes as well as some phenol and certain basic compounds, such pyridine, and is named by the fact that it is lighter than water. In order to eliminate bases and excess acid, the light oil is first washed with concentrated H₂SO₄ before being cleaned with NaOH. Three ingredients are recovered from the fractionally distilled, thusly purified oil:

Name of fraction	Temperature Range (°C)	Chief Constituents
90% Benzol	Up to 110	Benzene, toluene, xylene
50% Benzol	110 to 140	Benzene toluene xylene
Solvent Naphtha or Benzene	140 to 170	Xylene, Mesitylene

All the 3 fractions are read under their commercial names as water of coal, solvents and for dry cleaning, 90% Benzol is fractionated again to yield Benzene (800 °C), Toluene (1100 °C) and Xylene (1400 °C).

2. Following Fraction (Middle Oil, 170°C –230 °C):

Its name, carbolic oil, refers to the fact that it mostly contains naphthalene and carbolic acid. Naphthalene crystals are formed when it cools, leaving behind crude liquid carbolic acid. NaOH and dilute H₂SO₄ are used to purify naphthalene before it is finally sublimed. In order to dissolve the phenols, the crude carbolic acid is stirred with a NaOH solution. In order to extract free phenols, the solution is runoff and treated with weak H₂SO₄, this produces crystals of carbolic acid upon distillation. It is possible to create explosives, medicines, and colours with naphthalene. Phenol is used to create colours, antiseptics, explosives, bakelite and other products.

3. Third Fraction (Heavy Oil 230°C –270°C):

It is heavier than water. Its chief ingredients are carbolic acid and cresol. It is not purified further and is used as much under the name of cresol oil, for the preservation of timbers and for making disinfectants.

4. Fourth fraction:

"Green oil", which is primarily anthracene and ranges in temperature from 270°C to 360°C. When it cools, anthracene crystals, the raw ingredient for alizarin dyes, deposit themselves. Pitch is run out of the still while it is still hot. Still residue. It is used to create stoneware that is resistant to acids, tarred paper, roofing, and road surface. It also contains 92-94% carbon.



FACULTY'S ARTICLE

Corrosion inhibition by superhydrophobic surfaces



Prof. Devina Ratnam

Assistant Professor, B.I.T. Sindri

The most widely used designed material that is essential to many applications, from industrial equipment to building construction, from marine to aeronautical uses, is metals and alloys. Metals are extremely useful because of their wealth in intrinsic qualities, including great ductility, high mechanical strength but lightweight, malleability, and a high boiling point. However, rust restricts their use and lifespan by causing disintegration. Corrosion caused premature equipment failure, product pollution, fatalities, and financial loss. Given this ongoing problem, conventional techniques have been employed to prevent metal from degradation. However, the cathodic protection [2] approach is constrained by the high cost and widespread use of non-ferrous metals. Furthermore, an innovative approach to prevent corrosion in strong corrosive settings is to coat metal surfaces with an anti-corrosion layer.

Superhydrophobic surfaces (SHS) are highly water-repellent, and their capacity to self-clean has prompted researchers to learn more about this intriguing platform for preventing corrosion [5]. Polar water makes the least amount of contact with the surface, thanks to SHS's capacity to trap air in a micro or nanotextured surface. The surface's superhydrophobicity shields it from electrochemical deterioration as well.

Coatings have been bio-mimetic as a means to build a noble solution to corrosion. These coatings were inspired by nature, including lotus leaves, taro leaves, water striders, etc. They display contact angle hysteresis (CAH) of 10 and water contact angle (WCA) >150 . Because they can clean themselves, these surfaces have received a lot of attention due to characteristics that prevent icing, fouling, reduce drag, and prohibit corrosion.

Bioinspired, artificial SHS can be attained by two fabrication approaches:- i) surface modification by applying the low surface energy material, and ii) surface modification by introducing the hierarchical roughness on low surface energy material. Thus, surface energy and surface topography both contribute to the wetting and dewetting aspects of coating. Thus, the superhydrophobic coating acts as a corrosion barrier for Al, Cu, Mg, and steel [6] under seawater (NaCl solution) [2], basic, and acidic conditions [7]. Strong acids and bases can be resisted mechanically by TiO_2 , SiO_2 , and rare earth metal oxides. To create a superhydrophobic nanocomposite covering, Peng et al. combined epoxy resins with low surface energy fluoropolymers and implanted polytetrafluoroethylene (PTFE) nanoparticles [8]. It is discovered that the static water contact angle (WCA) is 158° and the sliding angle (SA) is 3° . It was discovered that the produced coating was stable for 6 months of storage in a room. The tape peel test and abrasion test both verified the coating's mechanical workability. Its WCA falls to 155° after 30 cycles, whereas SA underwent a modest shift of 1. Aqua regia, a highly concentrated solution of hydrochloric acid (HCl) and nitric acid (HNO_3) in a 3:1 volume ratio, and a 1M NaOH solution have both been used in experiments to test the changed substrate's chemical susceptibility. WCA only slightly changes after exposure to aqua regia and basic solution for 6 hours and 12 hours, respectively, but is still maintained at $>150^\circ$. These findings demonstrate its chemical resistance even in an extremely caustic environment. Halloysite nanotubes (HNTS) have been studied by Jiang et al. to create vibrant, wear- resistant superhydrophobic coatings [9].

Its main shortcomings, including poor durability [10] and reduced abrasion resistance SHS, restrict its use. The coatings lose their superhydrophobic properties shortly after application due to the experimentally discovered sensitivity to impact and attrition. Durability is therefore absolutely necessary for mechanical application in big machines. As a result, it is impossible to totally halt this problem, but it is more practical and affordable to take preventive measures to stop the metal surface from corroding through SHS. Traditional corrosion inhibitors have been in use for a very long time. However, the development of self-cleaning superhydrophobic surfaces has opened up new research avenues, and achieving mechanical robustness and attrition resistance will not be sufficient to halt corrosion's loss and devastation.

Reference:

1. EE. Ebenso, "Corrosion inhibition of mild steel in IM HCl by D-glucose derivatives (no, March, pp, 1-17,2017).
2. Y FAN, Y, He,P, Luo, X. Chen, and B.Liu,"Applied surface electrodeposition process to fabricate corrosion-resistant superhydrophobic surface on carbon steel, vol. 368, pp. 435-442, 2016.
3. S. G. Interface, "Control of Super hydrophilic and," 2013.
4. K. O. Zamuruyev et al., "Continuous Droplet Removal upon Dropwise Condensation of Humid Air on a Hydrophobic Micropatterned Surface, 2014.
5. W. Al Zoubi and Y. G. Ko, "Enhanced Corrosion Protection Performance by Organic- Inorganic Materials Containing Thiocarbonyl Compounds," Sci. Rep., no. July, pp. 1-11, 2018.
6. M. Parveen, M. Mobin, S. Zehra, and R. Aslam, "L-proline mixed with sodium benzoate as a sustainable inhibitor for mild steel corrosion in IM HCl: An experimental and theoretical approach," Sci. Rep., no. November 2017, pp. 1-18, 2018.

7. D.Zhang, L.Wang, H. Qian, and X. Li, "Superhydrophobic surfaces for corrosion protection: a review of recent progresses and future directions," J. Coatings Technol. Res., vol. 13, no. 1, pp. 11-29, 2016.
8. C. Peng, Z. Chen, and M. K. Tiwari," mechanochemical robustness and liquid impalement resistance," Nat. Mater., vol. 17, no. April 2018.
9. X. Jiang, M. Tian, Y. Lei, and T. Li, "Fabrication of colourful wear-resistant super hydrophobic coatings based on chemical modified halloysite," Colloids Surfaces A, vol. 602, no. June, p. 125117, 2020.
10. Fayyaz et al., "Enhancement of mechanical and corrosion resistance properties of electrodeposited Ni-P-TiC composite coatings," Sci. Rep., pp. 1-16, 2021.





STUDENT'S ARTICLE

Sangita kumari
Chemical Engineering, 2021

MITACS RESEARCH SCHOLAR 2024 The University of Cape Breton, Canada

The history of all great work bears witness to the fact that no great achievement has ever been accomplished without the active or passive support of one's surroundings. Consequently, I can see how mentoring enhanced my project execution. I am deeply grateful to Professor Adango Miadonye Sir and Professor Mumuni Sir of Cape Breton University's Chemical Engineering Department in Canada for their invaluable guidance throughout the project's completion. Last but not least, I would like to express my heartfelt appreciation to my parents, professors, seniors, and fellow colleagues for their unconditional support. I also extend my gratitude to Mitacs, a research company in Canada, for their financial support.

The project I worked on was a study on the phase behavior and viscosity correction of petroleum mixtures.

STUDENT'S ARTICLE

Reducing the viscosity of crude oil for efficient pipeline transportation is of paramount importance in the oil industry. Initially, the viscosity of crude oil can exceed 1 million mPa-s, posing a significant challenge for recovery, production, and transportation. Various techniques, including heating, dilution, emulsification, and core annular flow, can reduce the viscosity of heavy oil.

The dilution method is widely used for transporting heavy crude oil and bitumen due to its ease of application. Organic solvents and light hydrocarbons are used as diluents to reduce the viscosity of heavy crude oil. In this research, the dilution approach was employed using toluene, Terranova crude oil, alcohols (methanol, ethanol, and propanol), Saudi oil, and their binary mixtures.

The viscosity of alcohols and toluene was measured at various temperatures (20, 25, 30, 35, and 40°C). Then, the viscosity of Terranova crude oil was measured by mixing it with different compositions of toluene at varying temperatures, resulting in a reduction in crude oil viscosity. However, the impact of diluents diminished with increasing concentrations. Toluene showed a greater impact on reducing viscosity compared to alcohols.

STUDENT'S ARTICLE

The Puttagunta et al. correlation, modified Puttagunta et al. correlation in terms of composition, and the Cragoe Equation, Lederer equation by Shu (1984) were used to predict the viscosity of mixtures of ethanol and methanol. The viscosity reduction parameter was used to compare the predicted viscosity with the measured viscosity, yielding overall values of 3.6364, 3.6364, and 18.3696.

Viscosity correction of petroleum mixtures is crucial for optimizing their flow characteristics in various industrial applications. Petroleum products, such as crude oil, bitumen, and refined fuels, often exhibit high viscosity, which can hinder transportation through pipelines and processing in refineries.

The increasing demand for oil has led to the development of large reserves of thick and heavy oil. However, transporting this oil through pipelines is challenging due to its thickness, resulting in problems such as high pressure, increased costs, clogged pipes, and lost production. Heavy oil is defined as having a certain level of thickness (API gravity of 20 or above and/or a viscosity of 100 cP). It contains more complex molecules than regular oil, which can become unstable and cause problems during transportation, leading to losses.

STUDENT'S ARTICLE

To address this challenge, researchers and engineers employ various methods to adjust viscosity. Commonly used methods for reducing the viscosity of heavy crude oil include thermal treatment, emulsification, core annular flow, and dilution with hydrocarbons and organic solvents. Thermal treatment involves heating the petroleum mixture to reduce viscosity. This method is effective but can be costly due to energy requirements and the need for specialized equipment.

Another method involves diluting the petroleum with lighter hydrocarbons or solvents. For instance, blending heavy crude with lighter oils like gas condensates or using solvents like toluene or n-hexane can significantly reduce viscosity while preserving the original properties of the petroleum.

Emulsification is another technique where the petroleum is mixed with water or surfactants to form an emulsion. Although this reduces viscosity, there are challenges associated with separating the components back into their original forms after transportation or processing.

Core annular flow (CAF) is a technique used to improve the flow of oil through pipelines. Rather than directly reducing oil viscosity, CAF creates a thin layer of water along the inner pipe wall.

STUDENT'S ARTICLE

This layer acts as a lubricant, making it easier for the oil to move through the pipeline. However, maintaining consistent core annular flow can be difficult, as the flow regime often changes, and maintaining stable CAF conditions can be complex and requires careful monitoring.

Researchers are highly interested in using solvents to dilute heavy crude oil due to its practicality. This method forms the foundation of our work. A light solvent, such as toluene, can be used to reduce viscosity without changing the original characteristics of the hydrocarbon, which is an advantage over emulsification. It can also be used in different locations regardless of the climate, unlike thermal methods



STUDENT'S ARTICLE

Alok Ranjan
Chemical Engineering, 2021

Exploring Sustainable Solutions with Maithon Power Limited and IIT(ISM), Dhanbad

I am overwhelmed with the invaluable knowledge and experience this project has brought so far. I sincerely acknowledge Maithon Power Limited and IIT(ISM), Dhanbad, for providing this incredible platform to learn and explore. This journey would not have been possible without the constant support of my family, professors, mentors, and colleagues. I am especially grateful to the entire mentorship committee for their continuous guidance and encouragement.

The project I am working on is titled "Converting Plastic Waste into Sustainable Materials Using Pyrolysis". The growing concern over the adverse environmental impact of plastic waste demands innovative solutions. This project focuses on addressing the challenge by converting plastic waste into a valuable resource.

The process involves the pyrolysis of plastic waste, which breaks it down into hydrocarbons. These residues are then combined with carefully selected components such as natural fibers, biopolymers, and recycled materials to create a sustainable composite.



STUDENT'S ARTICLE

This material exhibits enhanced mechanical, thermal, and chemical properties, making it suitable for applications in construction, packaging, and various industries. Preliminary studies have shown promising results, with the composite offering improved durability and strength while adhering to sustainability goals. Additionally, the process emphasizes energy efficiency and employs a closed-loop waste management system to minimize the environmental footprint.

This mentorship program has provided me with a collaborative learning environment and the opportunity to interact with researchers, industry experts, and stakeholders. Through technical sessions, workshops, and practical insights, I have gained a deeper understanding of sustainable material development and its potential to contribute to a circular economy.



Students' Blogs



USE OF NANOPESTICIDES IN MODERN DAY AGRICULTURE

By ASHISH KUMAR SAHA

Batch 2021

Department. of Chemical Engineering, B.I.T Sindri

Without harming the biodiversity in the area, nanotechnology has provided excellent alternatives for insect pest management in agriculture, allowing for a more harmonious environment.

Nano Pesticides are nanostructures used to transport agricultural components that have two to three dimensions and range in size from 1 to 200 nm. When opposed to free insecticides, the loading of agrochemical ingredients into nanoparticles offers advantages due to their special features. However, a new category of environmental trash is being generated due to the rapid development of novel designed nanoparticles for pest management. Innovative nanopesticides are nanoparticles created to protect plants, minimize application losses, improve stability, boost leaf coverage, and decrease the amounts of formulation components. Formulations for nanopesticides can be split into active encapsulating components such nano emulsion, polymeric nanoparticles, lipid nanoparticles, and nanotubes, as well as self-organized systems like liposome, dendrimers, metallic and bimetallic nanoparticles.

Production has also been negatively impacted by the indiscriminate use of pesticides applied against harmful insects and pests. This has led to increased environmental imbalance, disease and insect resistance. This increased need for new agrochemicals to ensure the protection of the crops from various pests (pathogens, harmful insects, parasitic weeds) thus increasing production and productivity.



Students' Blogs



Benefits of Nano pesticides:-

To strike a dynamic balance between agricultural production and environmental sustainability, nanotechnology most likely offers a novel platform. Numerous studies have demonstrated that the application of nanopesticides in agriculture can significantly reduce costs, enhance shelf life, and improve nutritional value of agricultural products.

They are designed specifically to increase the solubility of insoluble or poorly soluble active ingredients and to release the biocide in a controlled and targeted manner, in contrast to conventional pesticide formulations. As a result, the application only requires a smaller amount of an active ingredient per area. Producing costs, non-target effects, and phytotoxicity are all reduced as a result of the reduced dose. Additionally, it is essential that controlled-release formulations remain inactive until the active in-gradient is released.

For e.g.:- Different nanopesticides like atrazine, that is mostly sprayed on sugarcane and maize crops, can induce DNA damage by a chemical reaction with adenine and guanine bases of the DNA.



Students' Blogs



The lunch box approach:-

The lunch box approach is mainly inspired from the mechanism of the "fake promises" and to attract the prey such as, predator spiders providing same sex signals as of moth to attract them and making of silk wrapped gifts by male spiders in order to mate with the female spider. Nanopesticides should, ideally, kill a target without harming other species or organisms by luring it. By combining a semio-chemical with a nano-delivery system, the nano lunch-box strategy eliminates the described randomness of encountering the pesticide.

It consists of three main steps:-

- (1) the lure
- (2) the box and
- (3) the kill.

The lure:-

The lunch-box method requires that the pest organism be seduced into this "belief" that it perceives an advantage in locating or being close to the "lunch". According to studies pheromones can be embedded in nanogels or polymer fibers by anchoring highly appealing chemicals on the surface of a nanocarrier. Semiochemicals, are the volatile compounds that indicate attraction and mating, food, or more broadly host- detecting chemicals, are potential attractants. Even at very low concentrations, these chemicals can be detected by organisms and induce a response that overrides many of an organism's natural "fears".



Students' Blogs



The box:-

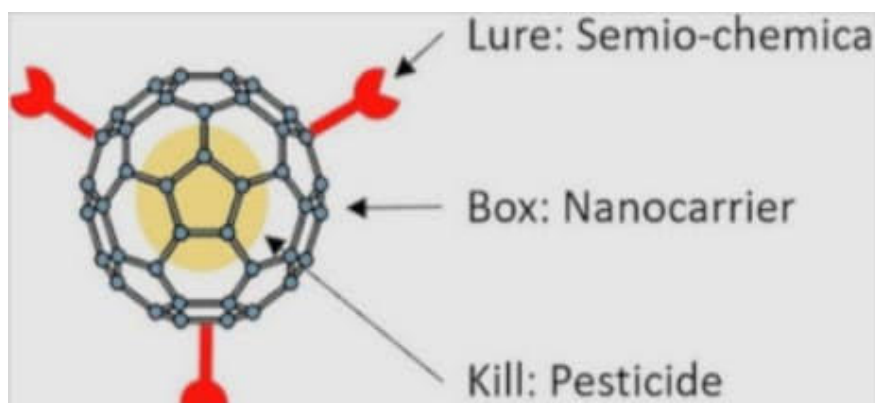
In this method, if a pest organism is drawn to the lunchbox, it must "open". The carrier must be made of something that can be eaten or digested, like cellulose or pectin. Gut digestive enzymes or physical-chemical factors may initiate the "opening". The stability of the carrier, its ability to be opened, and especially the timing of the box's opening is all dependent on the material. properties. Numerous natural materials can be broken down by enzymes found in living things, making them suitable for use in nanocarrier systems. As a result, site- specific pest control could get benefitted from their promotion of the release of active ingredients. Drug delivery and food science research served as inspiration for the majority of these site-specific release systems, whereas agricultural release applications-promoting systems are still in their infancy. Site-specific nanoparticle-based pesticide delivery methods are very interesting because they allow for precise effects on an organism while avoiding ineffective release.

The kill:-

The pesticide can begin working at the intended location without harming any other organisms once the lunchbox is opened. The method allows for the use of more gentle and sophisticated methods in addition to conventional chemicals. For example, Bt (*Bacillus thuringiensis*) can be used against various pest species, within the nanocarriers, novel natural or biosynthetic "compounds" can also be employed, e.g., natural chemicals, small-molecule agonists, or novel synthetic RNAi virus like strings.



Students' Blogs



Because the carrier system can protect and ensure proper functioning, the pesticides are more precise and generally cause less harm than conventional pesticide chemicals when used with this system. The "lunchbox" may even be designed to target specific tissues prior to release because nanocarriers can cross the midgut membrane however, development of this feature may take longer. It is possible to adjust what can be contained within the carrier as well as to enhance or inhibit cellular internalization by controlling the size of the carrier, such as between the nano and micro sizes. In conclusion, the lunch-box concept appears to be extremely promising for the creation of precision nanopesticides that enable targeted release, increased efficiency, and the avoidance of widespread pesticide-related negative effects. The interaction between ecological, nanotechnological, and chemical sciences benefits the approach.



Students' Blogs



GREEN HYDROGEN - A BOON

By SAHITYA SUMAN

Batch 2020

Department of Chemical Engineering, B.I.T Sindri

We know that due to our heavy reliance on fossil fuels, we produce over 830 million tons of carbon annually, which contributes to global warming, and we want to control/reduce this. For years, scientists and technocrats have been working to develop alternative fuels. The most recent studies by a group of scientist delegates from 195 countries have revealed the worst climate situation, particularly for Asian countries. The upcoming 26th UN Conference of Parties on Climate Change (COP 26) which is scheduled to take place on November 11, 2021, will make action plans to control greenhouse gases and climate change.

- Hydrogen is one of the most abundant elements on Earth, but we require the purest form of hydrogen, which is rare.
- On comparing the energy density of diesel with that of hydrogen, it is found that the energy density of hydrogen is three times that of diesel.
- 'Green Hydrogen' is a zero-carbon fuel. We can make this fuel by splitting water into oxygen and hydrogen using renewable energy from the sun and wind, a process known as electrolysis.



Students' Blogs



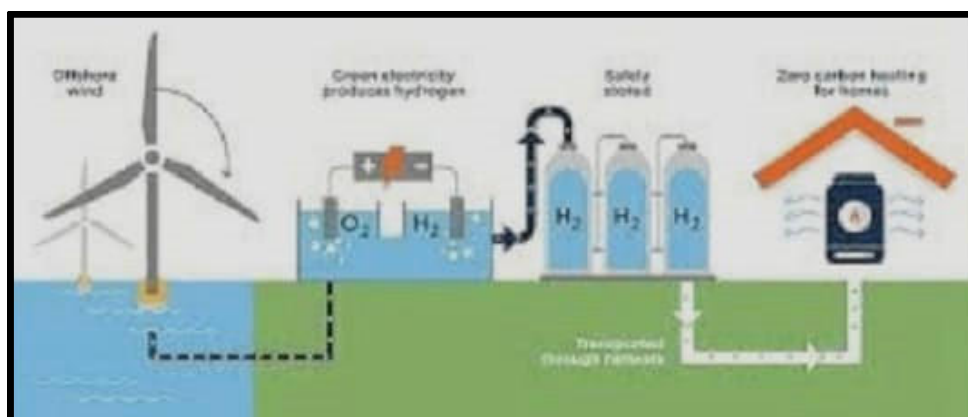
It is best to keep the temperature below 1.5 degrees Celsius. According to the International Energy Agency (IEA), the additional power demand will be between 25% and 30% by 2040.

- To meet the goal of expert guidelines on global warming of less than 1.5 degrees Celsius, power generation with 'net zero' emissions will be the best solution.

Challenges:-

Technical: The main issue is that in order to compress or liquefy the LH₂ (liquid hydrogen), it must be kept at a constant temperature of minus 253 degrees Celsius (which is much lower than the temperature of minus 163 degrees Celsius at which liquefied natural gas (LNG) is stored).

The cost factor: According to studies by the International Renewable Energy Agency (IREA), the manufacturing cost of a green source of energy is nearly 1.5 dollar per kg (for countries with constant sunlight and vast unused land) by 2030, taking various conservative measures into account.





Students' Blogs



What does Green Hydrogen Mean?

- Green hydrogen is a zero-carbon fuel produced by electrolysis using renewable wind and solar energy to split water into hydrogen and oxygen.
- This 'Green Hydrogen' could be used to generate power using natural resources such as wind or solar systems and would be a critical step toward achieving the goal of 'net zero emissions. In the current scenario, less than 0.1 percent, or 75 million tonnes of hydrogen per year, is produced, generating 284 GW of power.

More About Hydrogen:-

- Black hydrogen is produced using fossil fuels, whereas pink hydrogen is produced using electrolysis but with energy from nuclear power sources.
- Brown hydrogen is produced using natural gas, and the emissions are released into the atmosphere.
- Grey hydrogen is produced using natural gases, and the emission is released into the atmosphere.
- Blue hydrogen is produced using natural gas, and the emission is captured using carbon capture and storage.

Green Hydrogen, a thriving new concept, is a zero-carbon fuel produced by electrolysis using renewables from solar and wind to separate water into oxygen and hydrogen.

The issue with green hydrogen:-

- One of the major challenges that the industry faces in commercializing hydrogen or green hydrogen is economic sustainability.
- For transportation fuel cells, hydrogen must be cost-competitive with conventional fuels and technologies on a per-mile basis.
- Green hydrogen production is prohibitively expensive, and the necessary infrastructure is unavailable.
- Fuel cells, which convert hydrogen fuel into usable energy for automobiles, are extremely expensive.
- The infrastructure for hydrogen stations is woefully underdeveloped.

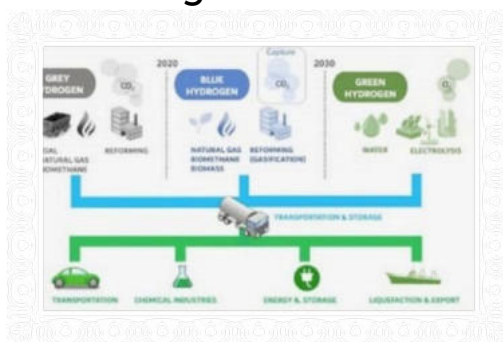


Students' Blogs



Indian Plans:-

- The India Railways has revealed that the country's first trial of a hydrogen fuel cell technology-based train will run under Northern Railway on the 89-kilometer stretch between Sonapat and Jind by retrofitting an existing diesel engine.
- The project will not only save a few lakhs in diesel engine costs each year, but it will also reduce the release of 0.72 kilotons of particulate matter and 11.12 kilotons of carbon per year.
- The National Hydrogen Energy Mission aims to produce hydrogen from natural energy resources.
- At the moment, India requires approximately 5.5 million tonnes of hydrogen, the majority of which is produced from imported fossil fuels.
- At the moment, India generates grey or blue hydrogen, which is produced using fossil fuels.



Conclusion:-

Hydrogen power generation will be a viable option for meeting the target of net-zero emissions while keeping global warming below 1.5 degrees Celsius. It will also be a significant step toward reducing reliance on conventional fossil fuels. It is past time for hydrogen to catch up with the rest of the world by embracing clean energy, decarbonizing the economy, and adopting 'Green Hydrogen' as an environmentally friendly and safe fuel for the next generation.

OUR ACHIEVERS



Vishwajeet Kumar, Shubham Kumar and Suraj Shaw got placements in the Maharatna Public Sector Undertaking IOCL.



This year, 2 students, Gorella Kumar and Vivek Pandey were the finalists in Sustain-A-Thon 2024.



OUR ACHIEVERS



Rajiv Mishra was shortlisted for BARC/TS.

Binit Kumar secured an All India Rank 50 in GATE-2024 and is now placed as a Graduate Engineer Trainee at the Maharatna Public Sector Undertaking IOCL.



OUR ACHIEVERS



RELIANCE INTERNS



ANSHU RANJAN
(2021-25)



ARUP PARAMANIK
(2021-25)



ARYAN DAS
(2021-25)



KUNDAN KUMAR
(2021-25)



MADHU KUMARI
(2021-25)



RAJ
(2021-25)



SHUVAM BANERJEE
(2021-25)



SUMAN KUMAR
(2021-25)



VISHWAJEET KUMAR
(2021-25)

OUR ACHIEVERS



TSU ISL



ALOK RANJAN
(2021-25)

MITACS GRI



SANGITA KUMARI
(2021-25)

TATA INTERNS



ANUPAMA KESHRI
(2021-25)



HEMANT KUMAR THAKUR
(2021-25)

IITs INTERN



ALOK MAHATO
(2022-26)
(IIT BHU)



KARTIK KUMAR
(2022-26)
(IIT BHU)



HARSHITA SETH
(2022-26)
(IIT BHU)



RAVI SHANKAR PRASAD
(2022-26)
(IIT DELHI)



PLACEMENT REPORT

Company:-

Aditya Birla Group

- | | |
|-----------------------|-----------|
| 1. Aneesha Liza Lakra | -2004009 |
| 2. Dinesh Kumar Yadav | -2104005D |
| 3. Md Junaid Alam | -2004039 |
| 4. Sakshi Kumari | -2004067 |
| 5. Swadha Shree | -2004083 |



Company:- Hindustan Unilever Limited

- | | |
|-------------------|----------|
| 1. Muskan Naaz | -2004042 |
| 2. Rachika Chawla | -2004052 |

Company:-

Indorama

- | | |
|-----------------------|----------|
| 1. Aditya kedia | -2004003 |
| 2. Madhu Kumari | -2004031 |
| 3. Ritik Kumar Thakur | -2004060 |
| 4. Roushan Kumar | -2004061 |
| 5. Sanjana Bharti | -2004070 |





PLACEMENT REPORT

Company:-

JSL

1. Kriti Kumari

- 2004098



Company:-

JSW

1. Abhinav Anand
2. Ashish Ranjan
3. Ashwani Kumar
4. Bikash Mondal
5. Harsh Singh
6. Kumar Gaurav
7. Kumar Satyam
8. Pappu Kumar Rajak
9. Sweta Swaraj

- 2004001
- 2004022
- 2004093
- 2004095
- 2004033
- 2004035
- 2004099
- 2004048
- 2004084

Company:-

JUSCO

1. Harsh Kumar Jha

- 2004096

Company:-

Larsen & Toubro

1. Anita Kumari

- 2004013





PLACEMENT REPORT

Company:-

Matix Fertilizers

1. Aryan Madhur
2. Pooja Kumari
3. Akash Kumar

-2004021
-2004050
-2004007



Company:-

Narayana Group

1. Mukesh Pandey
2. Nitish Kumar Mahato

-2104006D
-2004046

Company:-

Reliance Industry

1. Shubham Kumar
2. Vikash Kumar
3. Vivek Kumar

-2004077
-2004089
-2004039

Company:-

Tata Steel

1. Ankit Kumar Pandey
2. Nikhil Ranjan

-2004014
-2004044





PLACEMENT REPORT

Company:-

Shree Cement

- | | | |
|-------------------------|-----------|-----|
| 1. Anshu Kumari | -2004015 | • • |
| 2. Om Prakash | -2004047 | • • |
| 3. Sahitya Suman | -2004065 | • • |
| 4. Shubham Shekhar | -2004156 | |
| 5. Sudip Kumar Manjhi | -2004080 | |
| 6. Surendra Kumar | -2104012D | |
| 7. Vijaya Laxmi Hembrom | -2004088 | |

Company:-

Eastern Naptha

- | | |
|------------------------|-----------|
| 1. Rakesh Kumar Mahali | -2004056 |
| 2. Sourav Kumar | -2004079 |
| 3. Vikash Kumar Bhagat | -2004090 |
| 4. Adarsh Kumar | -2104001D |
| 5. Adarsh Kumar | -2004002 |

Company:-

Unacademy

- | | |
|------------------------|----------|
| 1. Avinash Kumar Singh | -2004094 |
|------------------------|----------|



PLACEMENT REPORT

Company:-

Vedanta Resources

1. Diksha Sinha	-2004029	• •
2. Preeti Kumari	-2004051	• •
3. Prerna Tirkey	-2004103	• •
4. Rashmita Soren	-2004057	
5. Shiwajee Ketan	-2004074	

Company:-

Technip Energies

1. Ashutosh Kumar Pandey	-2004024
2. Medha Singh	-2004040
3. Nikita Rani	-2004045
4. Shalvi Kumari Gupta	-2004072
5. Shruti Kumari	-2004105

Company:-

TCS- Ninja

1. Sanjeev Kumar Soren	-2104009D
------------------------	-----------



PLACEMENT REPORT

Company:-

TCE

1. Akash Pandey	-2004092	• •
2. Anima Khalko	-2004010	• •
3. Anish Kumar	-2004012	• •
4. Arjun Srivasata	-2004019	
5. Binit Kumar	-2004026	
6. Dhiraj Kumar	-2104013D	
7. Gorella Kumar	-2004031	
8. Gourav Kumar	-2004032	
9. Jitendra Kumar Ray	-2004097	
10. Nishant Kumar Yadav	-2004101	
11. Rahul Kumar	-2104008D	
12. Rishab Raj	-2004058	
13. Ritik Kumar Dubey	-2004059	
14. Shivam Kumar Mohali	-2104010D	
15. Shubham Kumar	-2004076	
16. Suraj Kumar Paswan	-2004082	
17. Vivek Kumar Pandey	-2004110	

Company:-

Dalmia

1. Avinav Raj Singh	-2004024
2. Shoaib Akhtar	-2004075



OUR RECRUITER





OUR RECRUITER



Hindustan Unilever Limited



LARSEN & TOUBRO



Shree
Cement



TATA CONSULTING ENGINEERS LIMITED

MATIX
FERTILISERS



unacademy



TATA
CONSULTANCY
SERVICES



TECHNIP
ENERGIES



Webinar >>>



Webinar on Career Guidance

Mr. Ankur Bansal, Head of Chemical Engineering at Unacademy Plus and an IIT Delhi graduate, delivered a webinar offering guidance on competitive exams like GATE, ESE, and CSE, along with insights on life skills and career planning.

Webinar >>>



Webinar on Profile Building and Placements

Miss Riya Singh and Mr. Birendra Kumar, 2016 graduates, conducted a webinar on profile building, resume writing, and strategies for placements and internships, offering valuable insights to attendees.

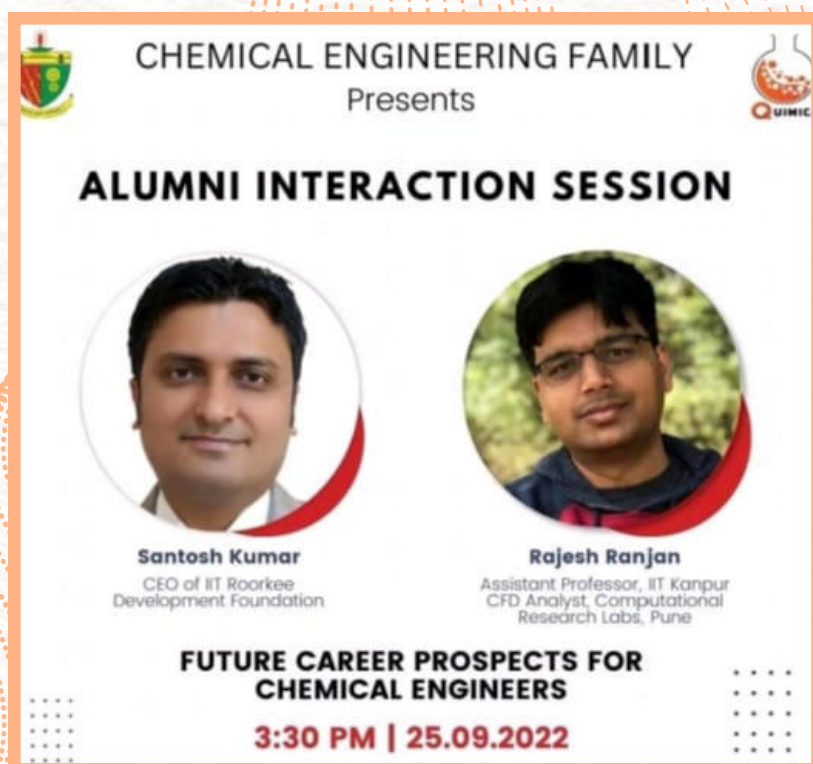
Workshop >>>



Placement Workshop

An online session with Mr. Saurabh Rai, Operations and Community Head and co-founder of Mentor Plus, covered key placement strategies, including recruitment frameworks and excelling in group discussions, offering valuable insights for placement readiness.

Seminar >>>



Alumini Interaction

An inspiring “Alumni Interaction Session” was held with distinguished alumni Dr. Santosh Kumar and Dr. Rajesh Ranjan. They shared invaluable insights from their personal journeys, highlighted the significance of education and revealed their mantras for achieving success.

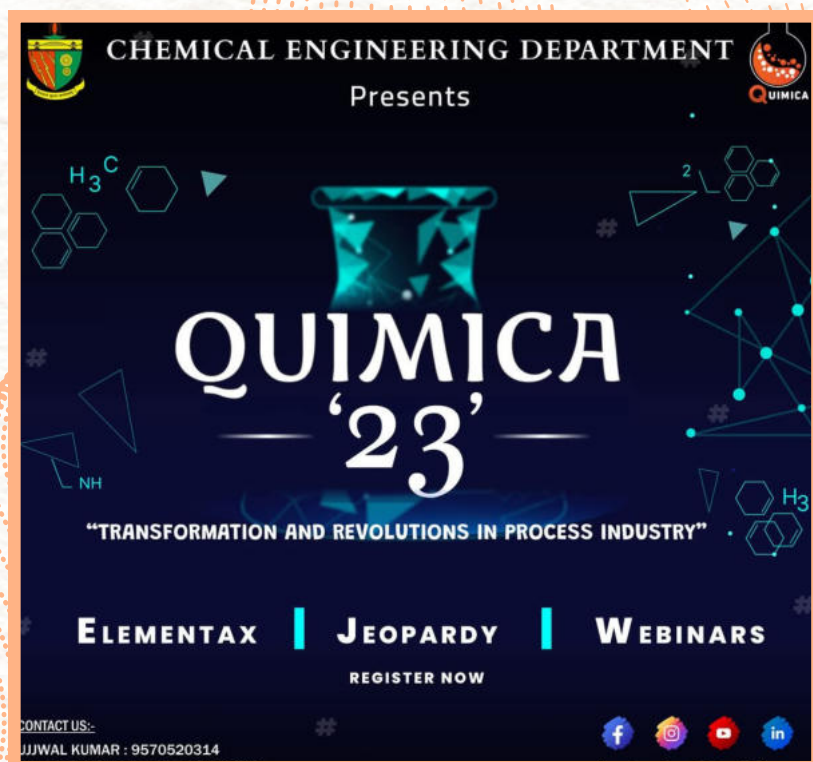
Seminar >>>



How to Get into MITACS?

MITACS-qualified seniors namely Miss Sakshi Gupta, Mr. Harsha Sinha, Miss Laxmi Sharma, Miss Mugdha Singh, and Miss Iysha Kumari answered queries regarding this Research Internship through the seminar. They answered all the questions about the Research Internship.

Recent Events >>>



QUIMICA'23

Quimica organises the annual event 'Quimica' which aims to encourage new minds to delve into academic studies and unveil the facts of Science and Engineering. QUIMICA'23 consisted of four Webinars and two events ElementaX and Jeopardy.

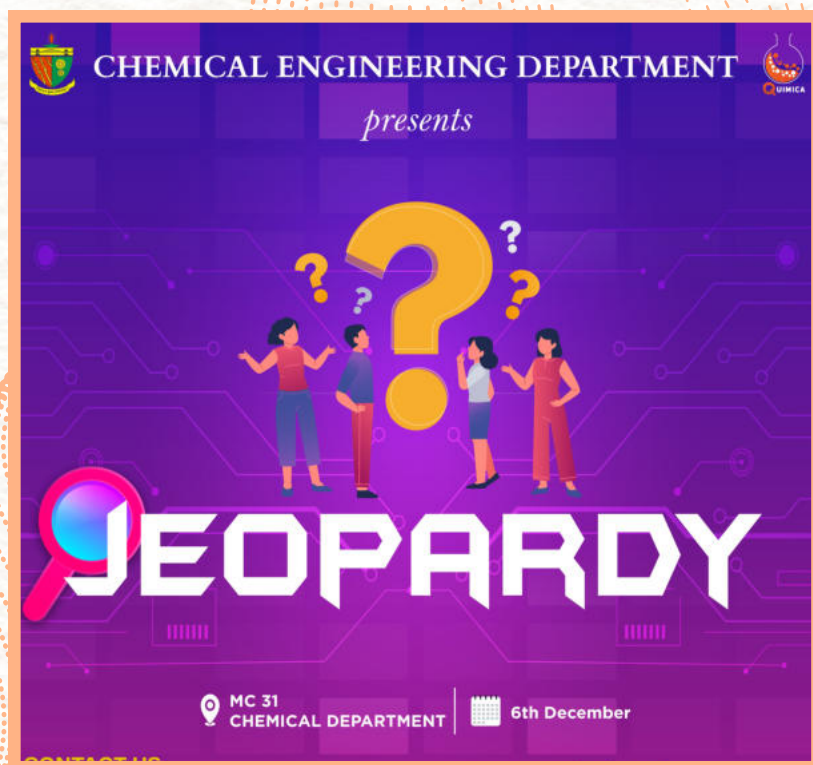
Recent Events >>>



ElementaX

ElementaX is an engaging event that encourages scientific exploration and academic growth through three rounds: a Crossword Puzzle, Presentation, and Q&A. Promoting collaboration and knowledge sharing, it's perfect for passionate learners competing for the title of ElementaX Champion.

Recent Events >>>



Jeopardy

The event “Jeopardy” is a dynamic and intellectually stimulating competition designed to test participants’ knowledge, creativity, and quick thinking. Featuring two engaging rounds—a pen-and-paper quiz and a Reverse Quiz Game, where participants create questions from given answers. By combining traditional quiz elements with an innovative twist, “Jeopardy” offers a unique and enriching experience.



Recent Events >>>



TECH-KRITI

TECH-KRITI was a dynamic technical event featuring a two-tier brainstorming competition with exciting challenges. In the first round, they selected a problem statement from a basket and presented innovative solutions as abstracts. The second round involved a face-off paper presentation before a panel of jurors, followed by a Q&A session based on the presentations.

Recent Events >>>



SCINTILLA

SCINTILLA focused on evaluating the pros and cons of chemical industry products, helping participants develop key debating skills for group discussions. The event encouraged critical analysis from diverse perspectives, allowing participants to showcase their creativity and innovative thinking. It fostered an environment of unconventional ideas, with students from prestigious universities across the country also taking part.

ALUMNI'S ARTICLE



Mr Nitish Kumar
Chemical Engineering, 2019

A biogas plant is to be set up near Hostel 26 to reuse the food waste from Hostel 7, Hostel 8, and Hostel 26. The biogas produced will be utilized in Hostel 26 mess in replacement of LPG. Mess will be charged as per the current market price of LPG/kg. The running of the plant and monthly collection of biogas bills from mess owners will be done by a professor in charge. Electricity bills, water bills, and plant maintenance will be done using the profit made by monthly collections. Urja Viable Solution (UVS) will promote the sustainable production of renewable energy from the biogas obtained from food and beverage industry waste (agro-food waste) in small-scale concepts for energy self sufficiency. This action will contribute to secure, sustainable, and competitively priced energy by promoting new and renewable energy sources and supporting energy diversification.

The project strategy is designed for the target group, the agro-food sector, which is at the same time the energy producer and the demand-side with the proposed scheme:

ALUMNI'S ARTICLE

First, the needs of end-users in terms of energy demand will be analyzed, as well as the difficulties they encounter when considering to install a biogas production facility.

Second, the necessary tools to address these needs will be developed according to the gathered information, including business collaboration models, small-scale process design and promotion, energy demand management models, and training to build-up of skills, awareness and networking.

Third, on-field actions will be implemented to promote this small-scale concept and to bring the developed tools to the end-users, including training sessions (face-to-face and on-line), workshops, webinars website, etc.

Specific programme of face-to-face activities with the most promising implementation sites will be carried out. UVS will produce a significant impact in terms of renewable energies uptake.

ALUMNI'S ARTICLE

It will act as catalyst and replication of success, triggering decisions to invest in renewable energy production from agro-food industry waste. Total transferability is guaranteed between participant countries that cover the most relevant regions of Hostels in terms of agro-food waste generation. The high relevance of the participants and their dissemination capacity will assure the high visibility of the BIOGAS - Sustainable small-scale biogas from agro-food waste for energy self-sufficiency 2 actions, making intelligent energy technologies as widely available as possible. A train-the-trainer approach ensures impacts beyond the actual project duration.

ALUMNI'S ARTICLE



Mr. Keshu Ranjan
Chemical Engineering, 2018

Quimica has provided me with a wonderful environment and opportunity to learn and grow myself academically as well as to secure a place in the corporate world. This is the time for all of you to understand the importance of any society with a vision of interacting more and more Chemical Engineers across the country. This college and my department has always guided and supported me for reaching my goal. The professors and seniors left no stone unturned to develop my communication and soft skills. Today, students from our department and our college are present almost in every corner of the world from various reputed institutions and research organisations to big shot corporate setups. Such efforts have inspired our current lot of students. These clubs and societies enhance the student's interactive skills on a professional level and help in personality development.

I always cherish the memories and the favourite pastimes I had while working as a part of the Quimica family. I feel immense felicity to have worked together as a team to convert every single opportunity. I always look forward to keeping in touch with all my fellow members. B.I.T. Sindri and my department will always be an inseparable part of my life for sure.

Recent Advancements in Chemical Industry

Blockchain: Wave of Substance in the chemical industry.

By: Saurabh Mishra, 2019 Batch.

Co-Convenor, Quimica.

Blockchain is a decentralized peer-to-peer digital network technology that can assist the chemical industry across its breadth, from building a coherent trading model to the development of client-centric products.

Trustless trust

In its "2018 Technology Vision", Accenture reported that 77% of chemical industry executives anticipated a timespan of one to three years for the integration of blockchain into company systems and 71% agreed to the gravity of blockchain in their company over the next three years. Largely known as the "trustless trust" setup, it shifts the trust factor from the participant to the system itself. The integral-cum-static design pushes further the exigency of blockchain integration in industry operations. Pioneers from various sectors viz. O&G, Chemical and Pharma, have started implementing this technology at different levels to avail industry-specific benefits like reducing workflow process, tokenizing the recycle value of its end products and legal improvements in its supply chain.

Recent Advancements in Chemical Industry

Nanotechnology for Water Purification.

By: Akash Deep, 2019 Batch

There is only 0.5% freshwater which is portable. The need is being felt to filter the impure water for human consumption. There are various technologies that we have encountered in our life like Reverse Osmosis (RO), UV filtration and many more. Scientists have discovered nanotechnology to purify water from contamination and unwanted minerals (RO) and make it portable. Nanotechnology is the process of manipulating atoms on a nanoscale.

Effectiveness

Nanotechnology is more effective than conventional ways like-Boiling, filtration, distillation, sedimentation, chlorination and oxidation. In this mechanism, two membranes are used for softening water; removing physical, biological and chemical impurities. This highly effective method has also been commercialized for safe-drinking water like Nanoceram, NanoH₂O, etc.

Recent Advancements in Chemical Industry

Main features

(a) More Surface and Small volume.

(b) Membrane material may change electrical, physical, chemical, optical, or biological properties at the Nano level.

(c) Chemical and biological reactions are easier. Under Nanotechnology, Nanocellulose based filtration is renewable and has high potential in water purification technology. Different materials are available such as Cellulose Nanocrystals (CNC) and Cellulose Nanofibrils (CNF). The Nanocellulose material is prepared by sulphuric acid hydrolysis and mechanical grinding method. The general water purification system is based on absorption. Though the nanotech method is highly efficient but has limitation in commercial production and incur high cost.

The field of nanotechnology in water purification is an important area of research. The degrading quality of fresh water for human consumption has made it a vital source for future water purifying systems.



Recent Advancements in Chemical Industry

Quantum Computing.

By: Anjali Shah, 2019 Batch.

The chemical industry is prepared to start to be an early beneficiary of the vastly expanded modeling and computational capabilities of quantum computing. Over the past years, quantum computing has been the subject of a lot of hype.

Work underway within the field at tech giants like IBM and Google has been extensively reported, and this interest has been mirrored by investments within the quantum-computing field by players from a wide range of industries including the chemical industry.

Adoption

This industry has been a comparatively late adopter of this digital innovation and practices moving across business and society. Quantum computing, uses the laws of quantum physics to extend the speed of certain calculations far beyond the capabilities of classical computers.



Recent Advancements in Chemical Industry

Future Aspects

For the industry, the new quantum-computing capabilities expose the likelihood of modeling quantum-mechanical systems, like molecules, polymers and solids, at a completely different level of precision. It would be possible to identify the most effective molecular designs or structures to accomplish specific tasks and achieve the required effects before synthesizing a single molecule in the lab.

Access to this kind of computation resource could exponentially boost the effectiveness of R&D departments and change the way new products are developed, with ramifications across the entire chemical industry

Nobel Prize in Chemistry



David Baker

David Baker, a distinguished professor at the University of Washington, was awarded the 2024 Nobel Prize in Chemistry for his groundbreaking work in computational protein design. In 2003, Baker successfully designed a novel protein unlike any found in nature, utilizing computational techniques he pioneered. This achievement opened the door to an immense diversity of new proteins, each tailored for specific functions.

Baker's work has profound implications across various sectors. In pharmaceuticals, his designed proteins have led to the development of more effective drugs and vaccines, enhancing disease treatment and prevention. In anomaterials, these proteins contribute to the creation of advanced materials with unique properties, driving innovation in technology and manufacturing.

Baker's achievements underscore the transformative potential of computational approaches in biochemistry, heralding a new era of protein engineering with far-reaching benefits for society.

Nobel Prize in Chemistry



Mouni G. Bawendi



Louis E. Brus



Aleksey Yekimov

In 2023, the Nobel Prize in Chemistry was awarded to Mouni G. Bawendi, Louis E. Brus, and Aleksey Yekimov for their pioneering contributions to the discovery and synthesis of quantum dots. These are tiny semiconductor nanoparticles that exhibit remarkable optical and electronic properties due to their nanometer-scale size. Quantum dots have revolutionized science and technology, unlocking numerous applications. Their unique ability to emit bright, tunable colors when exposed to light has been utilized in advanced displays, such as QLED televisions, providing sharper and more vibrant visuals.

The work of Bawendi, Brus, and Yekimov demonstrates the transformative potential of nanotechnology. By understanding and controlling materials at the atomic level, they have opened new avenues in fields like electronics, medicine, and quantum computing.

Nobel Prize in Chemistry



Demis Hassabis



John Jumper

Demis Hassabis and John Jumper, leading researchers at Google DeepMind, were awarded the 2024 Nobel Prize in Chemistry for their groundbreaking work in artificial intelligence. They developed AlphaFold, an innovative AI system that can predict the 3D structures of proteins with remarkable accuracy. This advancement has addressed one of biology's most challenging puzzles—understanding how proteins fold into their functional shapes. Knowing their precise structures is critical for understanding diseases, designing effective drugs, and creating novel biotechnological solutions.

The Nobel Committee recognized Hassabis and Jumper's achievement as a transformative leap for both chemistry and biology. Their work exemplifies the potential of AI in solving complex scientific problems, marking a new era in interdisciplinary innovation.



GALLERY







POST BEARERS FOR 2024-25

DESIGNATION

NAME (BATCH)

PRESIDENT

SURAJ KUMAR (2K21)

VICE PRESIDENT

VIKAS KUMAR (2K21), MADHU KUMARI (2K21)

SECRETARY

VISHNU VAIBHAV (2K21)

JOINT SECRETARY

RAVI KUMAR (2K21), ASHISH KR. SAHA (2K21)

TREASURER

SYED SHIFRAN AHMAD (2K21)

JOINT TREASURER

UJJWAL KUMAR (2K21), SHIV PRASAD MARANDI (2K21)

TECHNICAL HEAD

ABHAY ANAND (2K21)

DESIGN AND CREATIVE HEAD

ADITI KUMAR (2K21)

PUBLIC RELATION OFFICER

ABHAY KUMAR (2K21)

ALUMNI IN-CHARGE

AMISHA (2K21), NISHA KUMARI (2K21)

TECHNICAL TEAM

AASHI RANI (2K22), HARSH (2K22), KARTIK KUMAR (2K22), RAVI SHANKAR PRASAD (2K22), SOURAV KUMAR (2K22), AKASH NATH (2K22), MANOJ MURMU (2K22), SAHITYA KUMAR (2K22)

CONTENT DEVELOPMENT TEAM

DEEPTI KUMARI (2K22), DIKSHA (2K22), DONA BHATTACHARJEE (2K22), HARSHITA SETH (2K22), KOMAL KUMARI (2K22), NIDHISHREE MAHATO (2K22), RUDRANIL GANGULY (2K22), SOURAV OMONG (2K22), SAMRIDDHI SINGH (2K22)

MANAGEMENT & LOGISTICS TEAM

AKASH NATH (2K22), KARTIK KUMAR (2K22), RAVI SHANKAR PRASAD (2K22), RUDRANIL GANGULY (2K22), SAHITYA KUMAR (2K22), SOURAV KUMAR (2K22), SOURAV OMONG (2K22)



/quimicabits_blog

VISIT OUR SOCIALS



WITH OUR DEEPEST GRATITUDE...

WE'RE EXCITED TO INTRODUCE, QUIMICABITS, OUR OFFICIAL NEWSLETTER, WHICH WOULD HELP YOU ALL TO FOSTER A FEELING OF COMMUNITY AND ATTACHMENT TOWARDS OUR CHEMICAL DEPARTMENT.

WE OWE A HUGE DEBT OF GRATITUDE TO EVERYONE WHO PUT IN COUNTLESS HOURS TO CREATE THIS MAGAZINE AND MAKE IT A PERFECT WORK OF ART.

WE WOULD LIKE TO THANK OUR DIRECTOR PROF (DR.) D.K.SINGH SIR AND HOD CHEMICAL ENGINEERING - PROF (DR.) AMIT KUMAR GUPTA WHOSE EXCEPTIONAL TECHNICAL SUPPORT AND LOGISTICAL ASSISTANCE ACTED AS A BEACON OF GUIDANCE FOR US

THE GREATNESS OF OUR INSTITUTE AND PUBLICATION HAS BEEN EXCLUSIVELY ATTRIBUTED TO OUR DISTINGUISHED FACULTY, WHO HAVE MADE IT POSSIBLE FOR US TO PUBLISH THIS MODEL OF EXCELLENCE. WE WOULD LIKE TO SHOW OUR SINCERE GRATITUDE TOWARDS THEM.

ALSO, WE WOULD LIKE TO EXPRESS OUR GRATITUDE TO THE SHREWD AND INVENTIVE ALUMNI, WHO GAVE US HOPE THROUGHOUT THIS ROUGH ROAD, WHICH WOULD HAVE PREVENTED US FROM GETTING THIS FAR. WE SINCERELY THANK EVERY ONE OF THEM FOR THEIR EXTRAORDINARY WORK AND COMMITMENT.

ADDITIONALLY, WE WOULD LIKE TO SHOW OUR THANKS TO THE POST- BEARERS OF THE CHEMICAL ENGINEERING SOCIETY FOR INSPIRING US TO GO ABOVE OUR BOUNDARIES AND FOR HELPING US TO DEVELOP THIS PIECE OF EXCELLENCE. WE ARE LUCKY TO HAVE STUDENT COORDINATORS WHO ARE SO BRILLIANT AND COMPETITIVE AND WHOSE EXCEPTIONAL ACCOMPLISHMENTS ARE AN INSPIRATION TO US.

AFTER GOING THROUGH THIS MAGAZINE, YOU WILL COMPLETELY COMPREHEND THE POTENTIAL OF THE CHEMICAL ENGINEERING DEPARTMENT, WHICH YOU WILL ALL ADMIRE IN THE FOLLOWING YEARS.